Class: BE **Subject: Applied Data Science Lab** **Sem-VIII**

Experiment No 9

Course Outcome: CO5 **Blooms Level: L3**

Aim: Implement time series decomposition and moving averages method of trend estimation.

Abstract: Real-world data collected over time often exhibits patterns such as long-term trends, seasonal variations, and random fluctuations. Identifying these components is important for understanding the behavior of the data and for making accurate predictions. However, separating these components from the original time series can be challenging. Therefore, there is a need for methods that can analyze time-dependent data and identify its underlying patterns. Apply time series decomposition techniques to separate the dataset into trend, seasonal, and noise components. Estimate the trend component using the moving averages method and visualize the different components to better understand the underlying patterns present in the time series data.

Theory: A given time series is thought to consist of three systematic components including cycle, trend, seasonality, and one non-systematic component called noise.

In an additive decomposition, the components are decomposed in such a way that when they are added together, the original time series can be obtained.

Time series=Trend + Seasonality + Noise

In the case of multiplicative decomposition the components are decomposed in the such a way that when they are multiplied together, the original time series can be derived back.

Time series=Trend * Seasonality * Noise

Both additive and multiplicative time series decomposition can be represented by these equations:

where Tt, St, and Et are trend, seasonal, and error components respectively.

The first step in a classical decomposition is to use a moving average method to estimate the trend-cycle

Moving Averages Method gives a trend with a fair degree of accuracy. In this method, we take arithmetic mean of the values for a certain time span. The time span can be three-years, four -years, five- years and so on depending on the data set and our interest. We will see the working procedure of this method.

Program:

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
product = {'month' : [1,2,3,4,5,6,7,8,9,10,11,12],'demand':[290,260,288,300,310,303,329,340,316,330,308
,310]}
```

In [3]:
```
df = pd.DataFrame(product)
```

In [4]:
```
df.head()
```

Out[4]:

	month	demand
0	1	290
1	2	260
2	3	288
3	4	300
4	5	310

In [5]:
```
df['pandas_SMA_3'] = df.iloc[:,1].rolling(window=3).mean()
```

In [6]:
```
df.head()
```

Out[6]:

	month	demand	pandas_SMA_3
0	1	290	NaN
1	2	260	NaN
2	3	288	279.333333
3	4	300	282.666667
4	5	310	299.333333

In [7]:
```
df.head()
```

Out[7]:

	month	demand	pandas_SMA_3
0	1	290	NaN
1	2	260	NaN
2	3	288	279.333333
3	4	300	282.666667
4	5	310	299.333333

In [8]:
```
df['pandas_SMA_4'] = df.iloc[:,1].rolling(window=4).mean()
```

In [9]:
```
df.head()
```

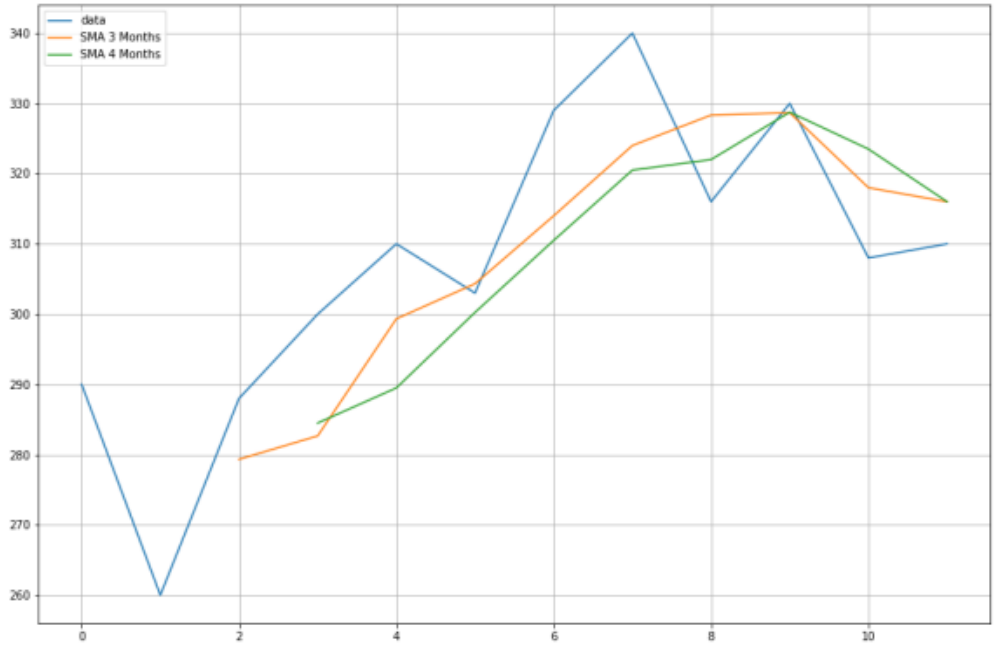
Out[9]:

	month	demand	pandas_SMA_3	pandas_SMA_4
0	1	290	NaN	NaN
1	2	260	NaN	NaN
2	3	288	279.333333	NaN
3	4	300	282.666667	284.5
4	5	310	299.333333	289.5

In [10]:
```
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [12]: plt.figure(figsize=[15,10])
         plt.grid(True)
         plt.plot(df['demand'],label='data')
         plt.plot(df['pandas_SMA_3'],label='SMA 3 Months')
         plt.plot(df['pandas_SMA_4'],label='SMA 4 Months')
         plt.legend(loc=2)

Out[12]: <matplotlib.legend.Legend at 0x1cb7beb2488>
```



Conclusion: Time series decomposition is one of the best ways to understand how a time series behaves. The statsmodels library provides an implementation of the naive, or classical, decomposition method in a function called seasonal_decompose().

Exercise 1: Time Series Decomposition (Car Retail Dataset)

Abstract:

Retail sales data collected over time often exhibits patterns such as long-term trends, seasonal variations, and random fluctuations, making it difficult to directly interpret the underlying behavior. Identifying and separating these components is essential for accurate analysis and forecasting. However, extracting these patterns from raw data can be challenging due to their overlapping nature. Therefore, there is a need for techniques that can decompose time series data into its fundamental components. Apply time series decomposition methods on the car retail dataset to separate it into trend, seasonal, and residual components, and analyze the patterns present in the data.

Exercise 2: Exponential Smoothing

Abstract:

In many real-world scenarios, demand data collected over time may exhibit fluctuations due to random variations, making it difficult to identify the underlying pattern. Traditional methods like moving averages assign equal weights to past observations, which may not adequately reflect recent changes. Therefore, there is a need for a method that gives more importance to recent data points. Exponential smoothing is a forecasting technique that assigns exponentially decreasing weights to older observations

while giving higher importance to recent values. This experiment focuses on applying Simple Exponential Smoothing (SES) to smooth demand data and generate forecasts. The results help in understanding how smoothing reduces noise and how future values are predicted based on past observations.

Dataset:
Monthly product demand dataset (given in program)

Exercise:
1. Implement Simple Exponential Smoothing on the given demand dataset using a smoothing constant $\alpha = 0.3$.
2. Compute the smoothed values manually for the first few observations.
3. Use Python to calculate smoothed values and forecast the next 3 periods.
4. Plot the original data, smoothed values, and forecasted values.
5. Analyze the effect of different values of αon the smoothing behavior.